DEVOPS INTRO

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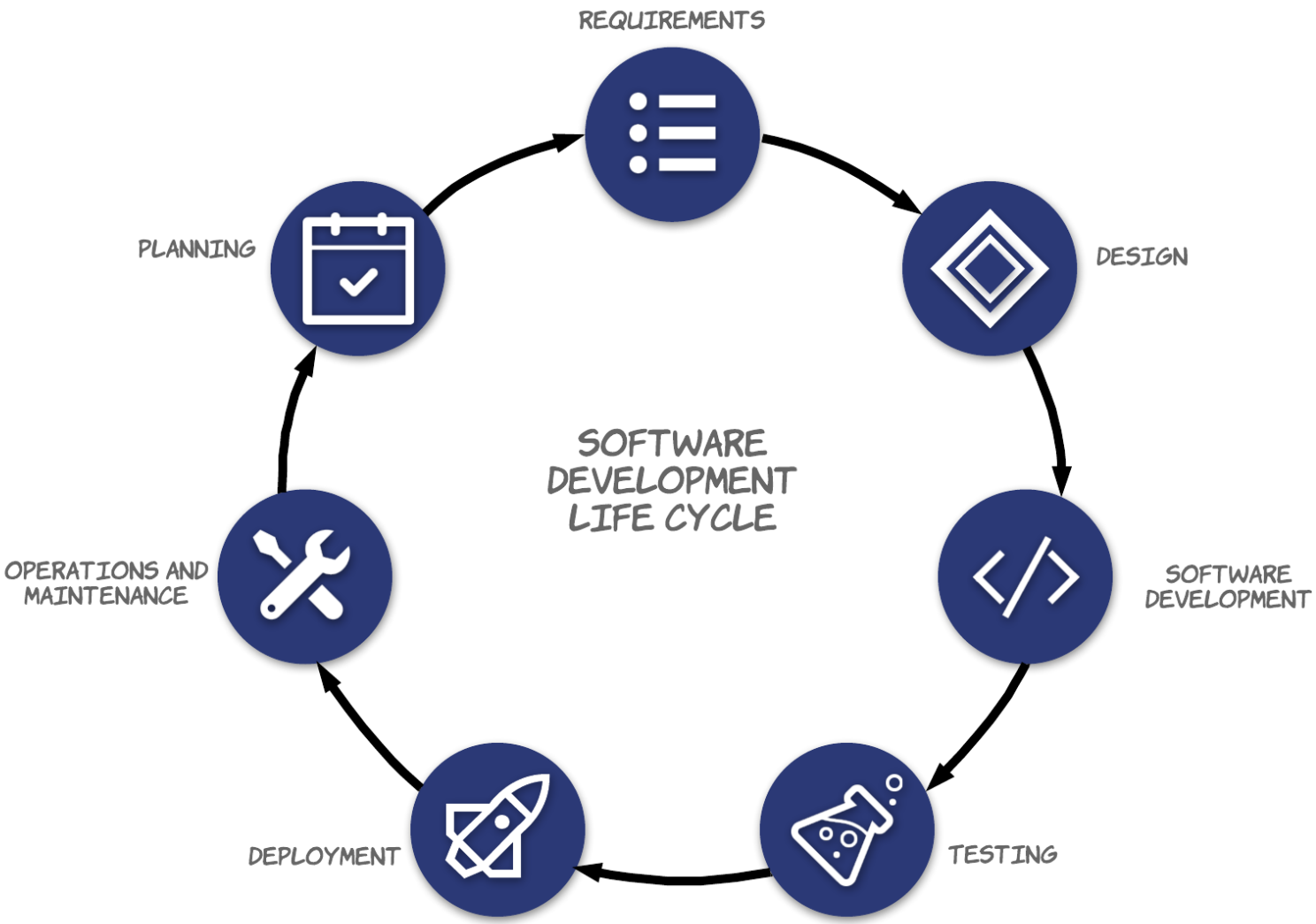
Lead Cloud Solution Architect

AGENDA

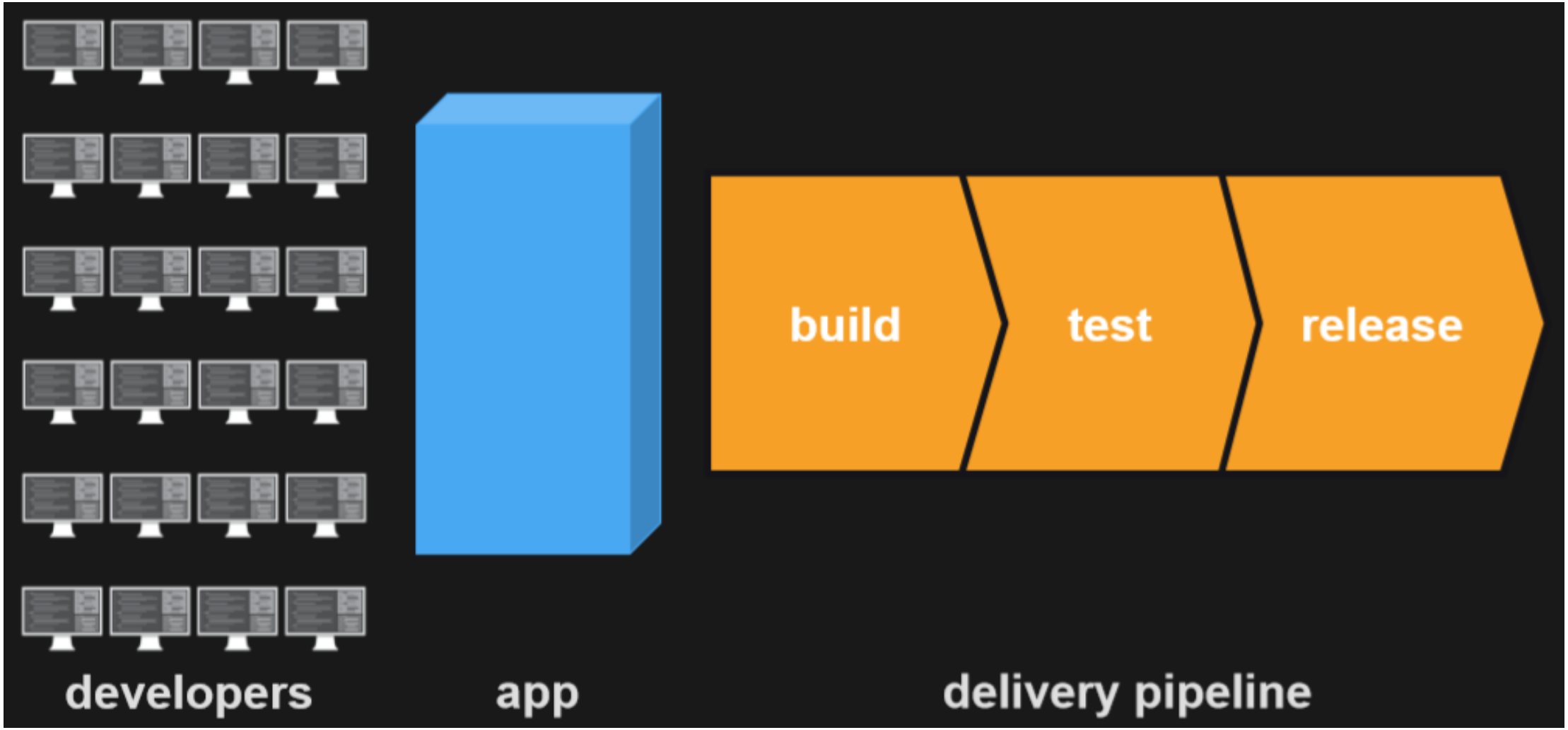
- Software Development Life Cycle / Agile
- What is Devops
- Why Devops?
- Devops vs Release Management
- When to adopt and when not to
- DevOps LifeCycle Phase
- Pipeline Overview

SOFTWARE DEVELOPMENT LIFE CYCLE / AGILE

SOFTWARE DEVELOPMENT LIFE CYCLE



MONOLITH DEVELOPMENT LIFECYCLE



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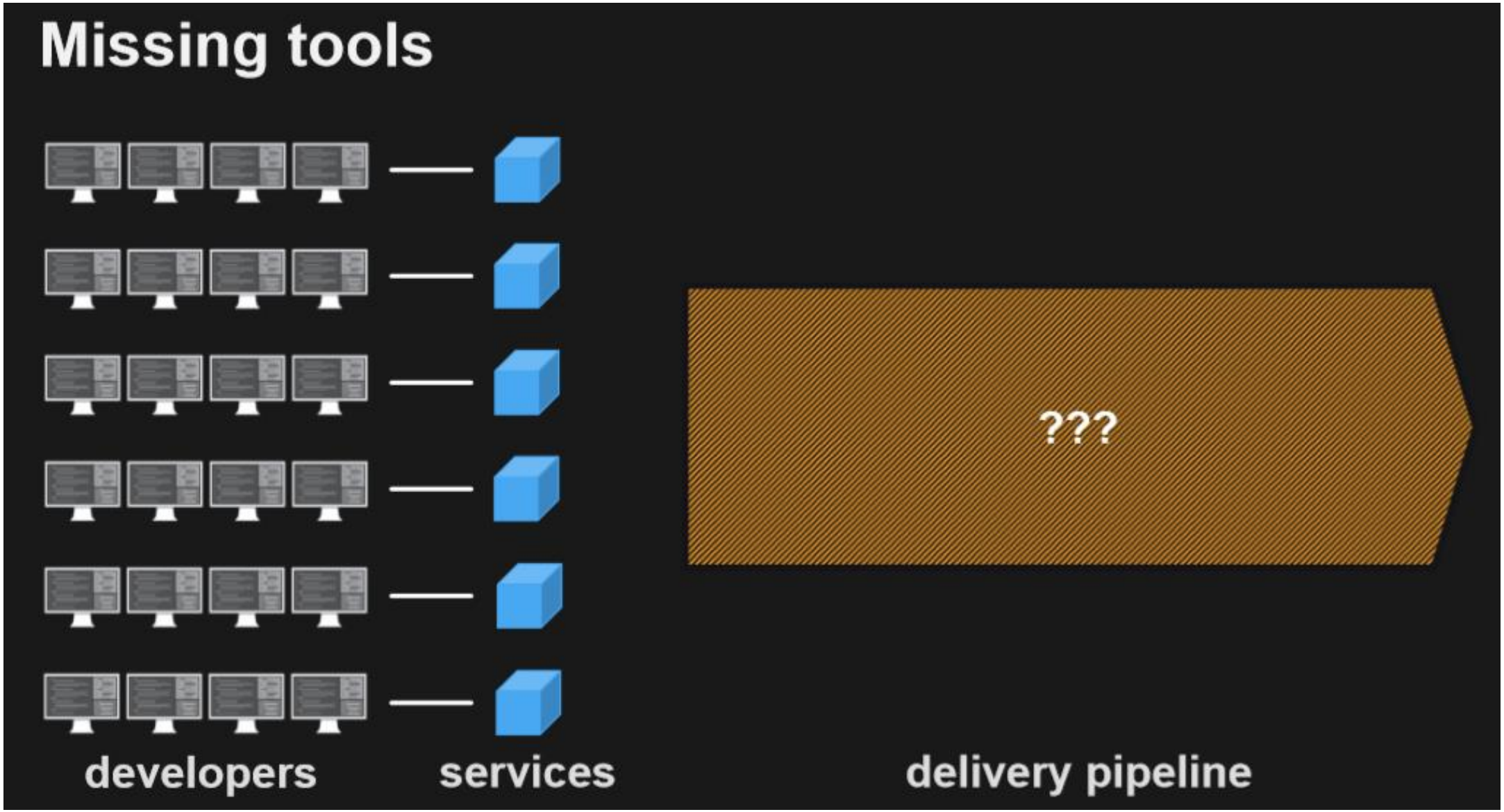
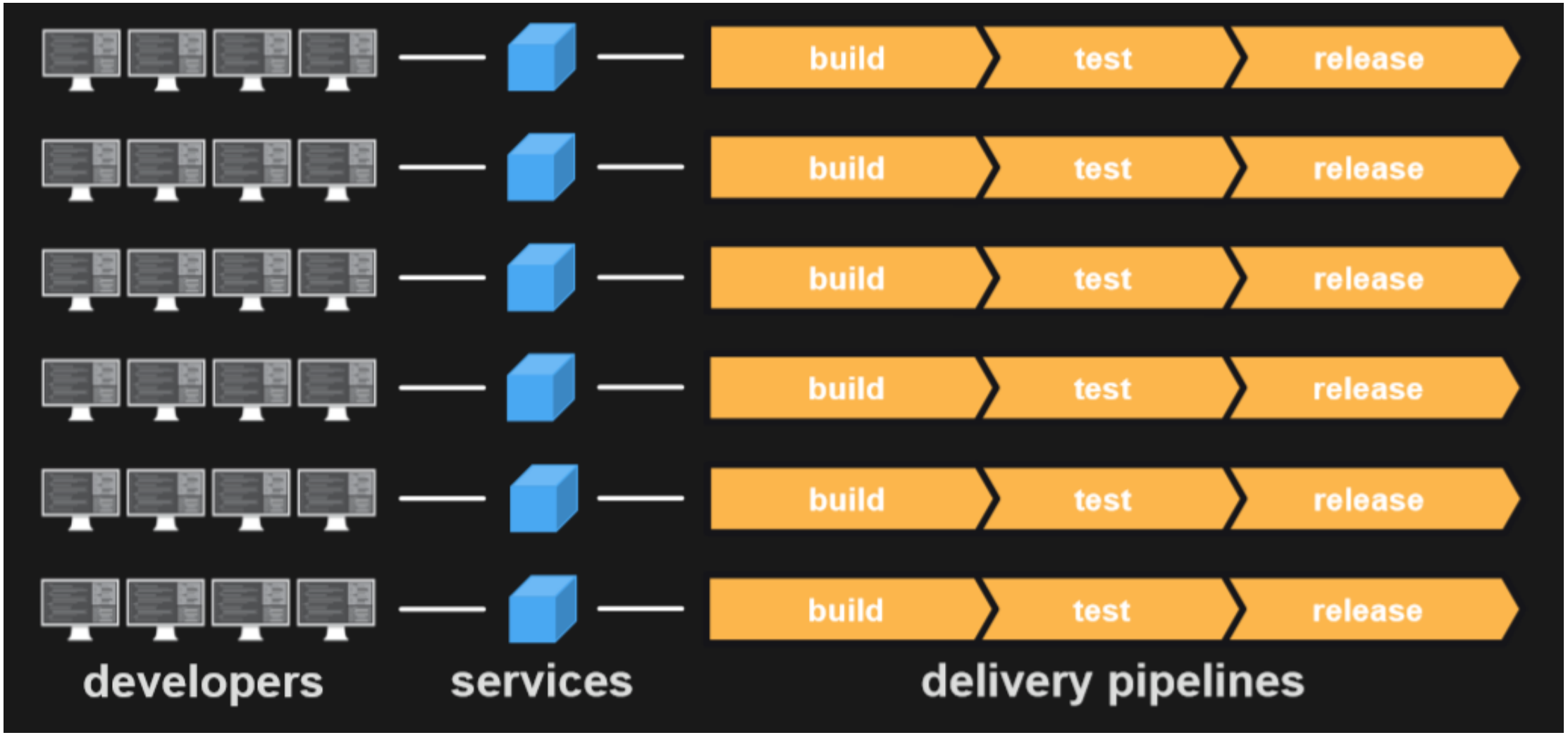


MICROSERVICE DEVELOPMENT LIFECYCLE



AGILE SOFTWARE DEVELOPMENT

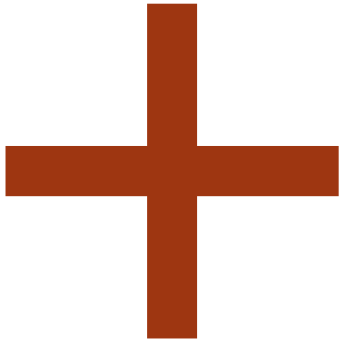
- **Agile software development** is a conceptual framework for software engineering that promotes development iterations throughout the life-cycle of the project.

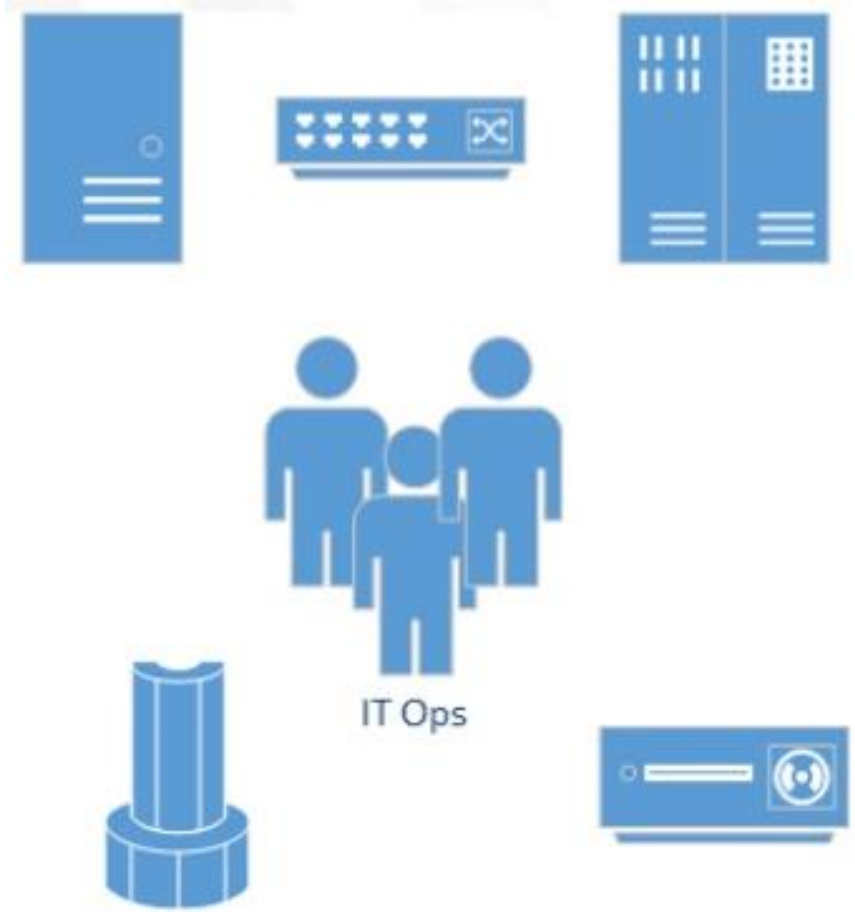
- Software developed during one unit of time is referred to as an iteration, which may last from one to four weeks.

- Agile methods also emphasize working software as the primary measure of progress

WHAT IS DEVOPS?

WHAT IS DEVOPS?
-- EFFICIENCIES THAT SPEED UP LIFECYCLE





WHAT IS DEVOPS?

- Devops is a software development method that stresses
  - Communication,
  - Collaboration
  - Integration

  Between Software Developers and Information technology professionals thereby

  Enable rapid evolution of products or services

  Reduce risk, improve quality and reduce costs

WHY DEVOPS?

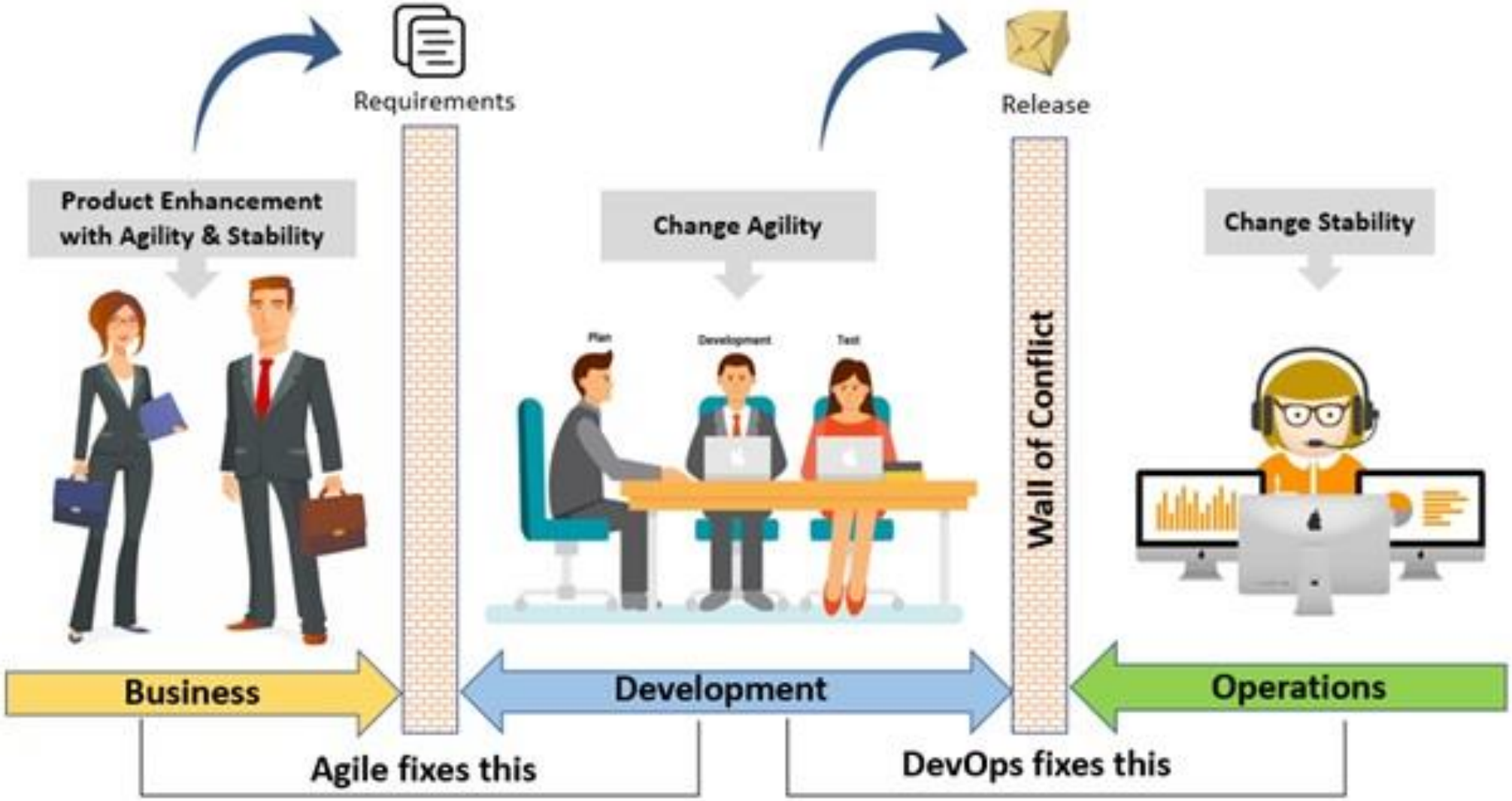
WHY GAPS?

- Developer View
  - Delivers features after testing in development systems
  - Dev system may not be same as production system
  - Developers will have faster turn around time w.r.t features
  - Not much concerned about the infrastructural as well as deployment impact because of the code changes

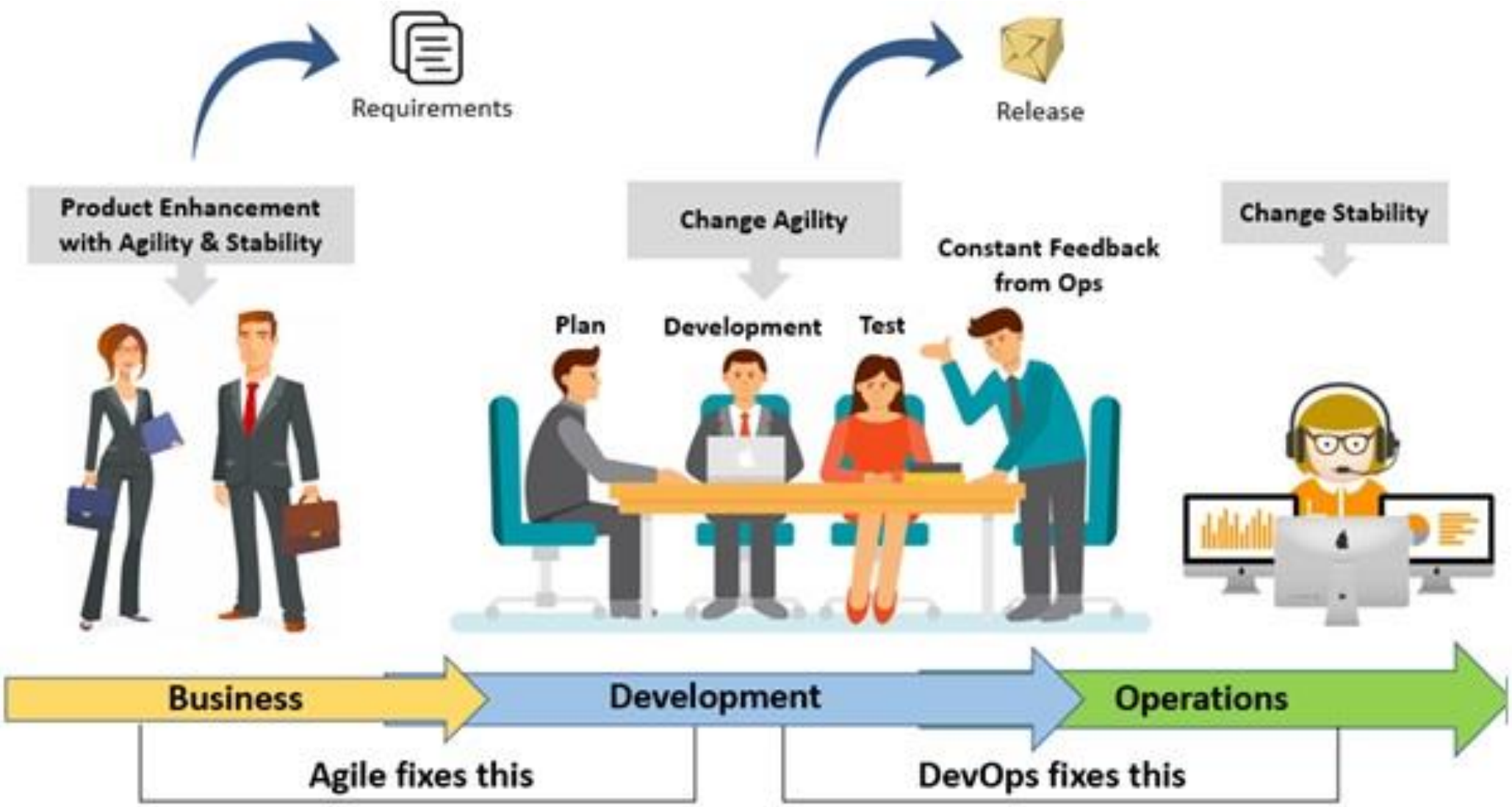
WHY GAPS?

- Ops View
  - Rewarded mainly for uptime
  - Lesser turn around time w.r.t feature deployment and testing due to large number of dev build coming their way
  - Very much concerned about the infrastructural as well as deployment impact because of the code changes

WHY GAPS?


Requirements
Release
Product Enhancement with Agility & Stability
Change Agility
Change Stability
Wall of Conflict
Business
Development
Operations
Agile fixes this
DevOps fixes this

DEV AND OPS


Requirements
Release
Product Enhancement with Agility & Stability
Change Agility
Change Stability
Constant Feedback from Ops
Plan
Development
Test
Business
Development
Operations
Agile fixes this
DevOps fixes this

DEV AND OPS

- Developers work with Ops to understand the impact of code changes
- Ops now have more clarity on infra needs
- Closely monitors the DEV – TEST – PROD pipeline for each deployment with immediate feedback

TOP 3 DELIVERY CHALLENGES ADDRESSED

- Release management
  - Better understanding of risks, dependencies, compliance issues
- Release/Deployment Coordination
  - Better tracking of discrete activities, faster escalation of issues, documented process control and granular reporting
- Release/Deployment Automation
  - Usually have existing automation but want to flexibly manage and drive this automation that can be invoked by non-operations resources for non-prod env.

DEVOPS VS RELEASE MANAGEMENT

WHEN TO ADOPT AND WHEN NOT TO

WHEN TO ADOPT AND WHEN NOT TO

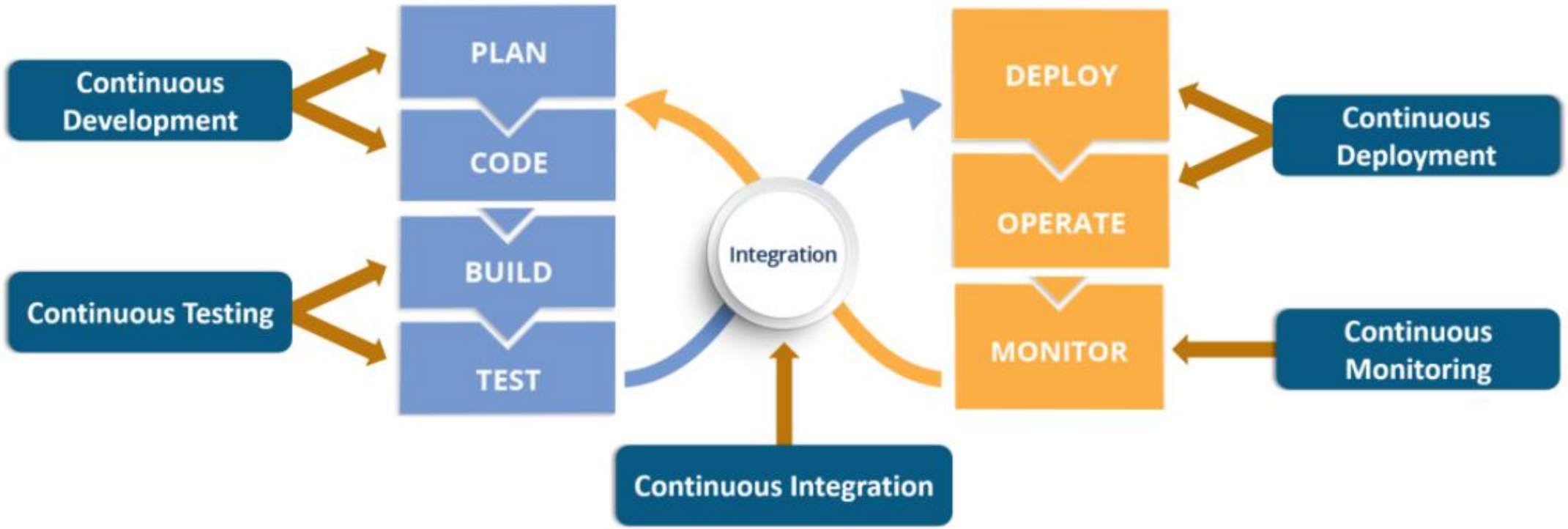
- When to Adopt:
  - For eCommerce and other web site projects (Amazon, Flickr, etc)
  - Cloud Platform (IaaS and PaaS)

- When NOT to Adopt:
  - Mission critical applications (Banks, Power systems etc)

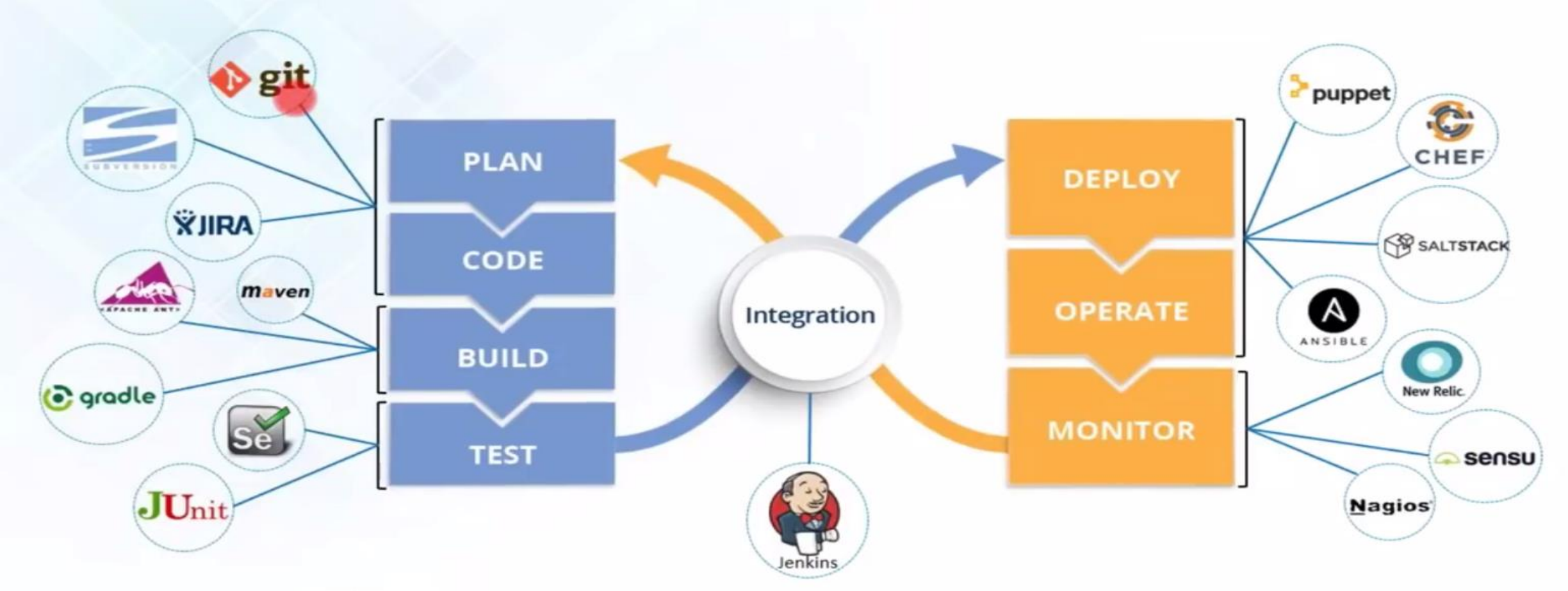
DEVOPS LIFECYCLE PHASE

DevOps Life Cycle Phases

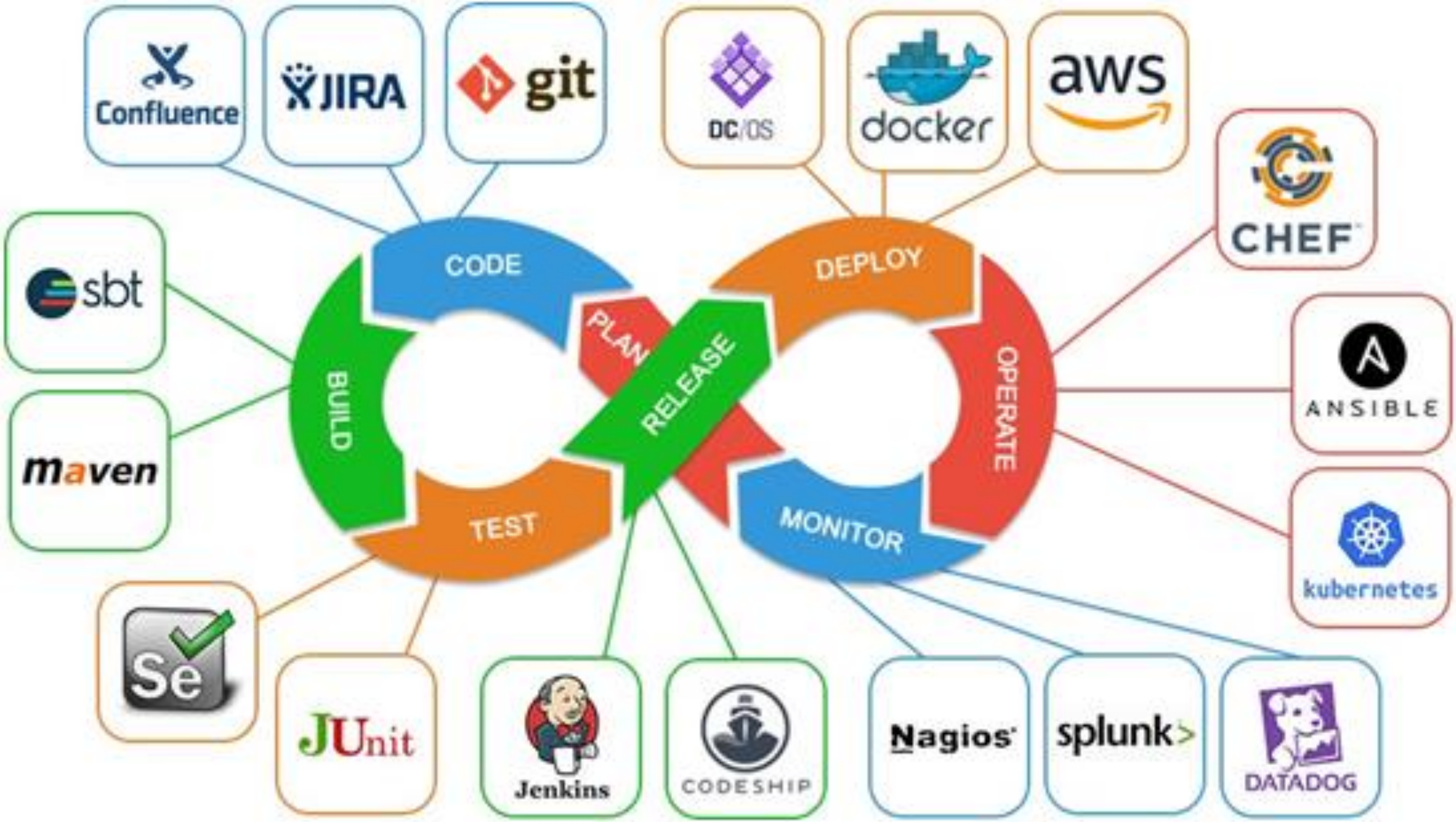
- Continuous Development
- Continuous Testing
- Continuous Integration
- Continuous Deployment
- Continuous Monitoring

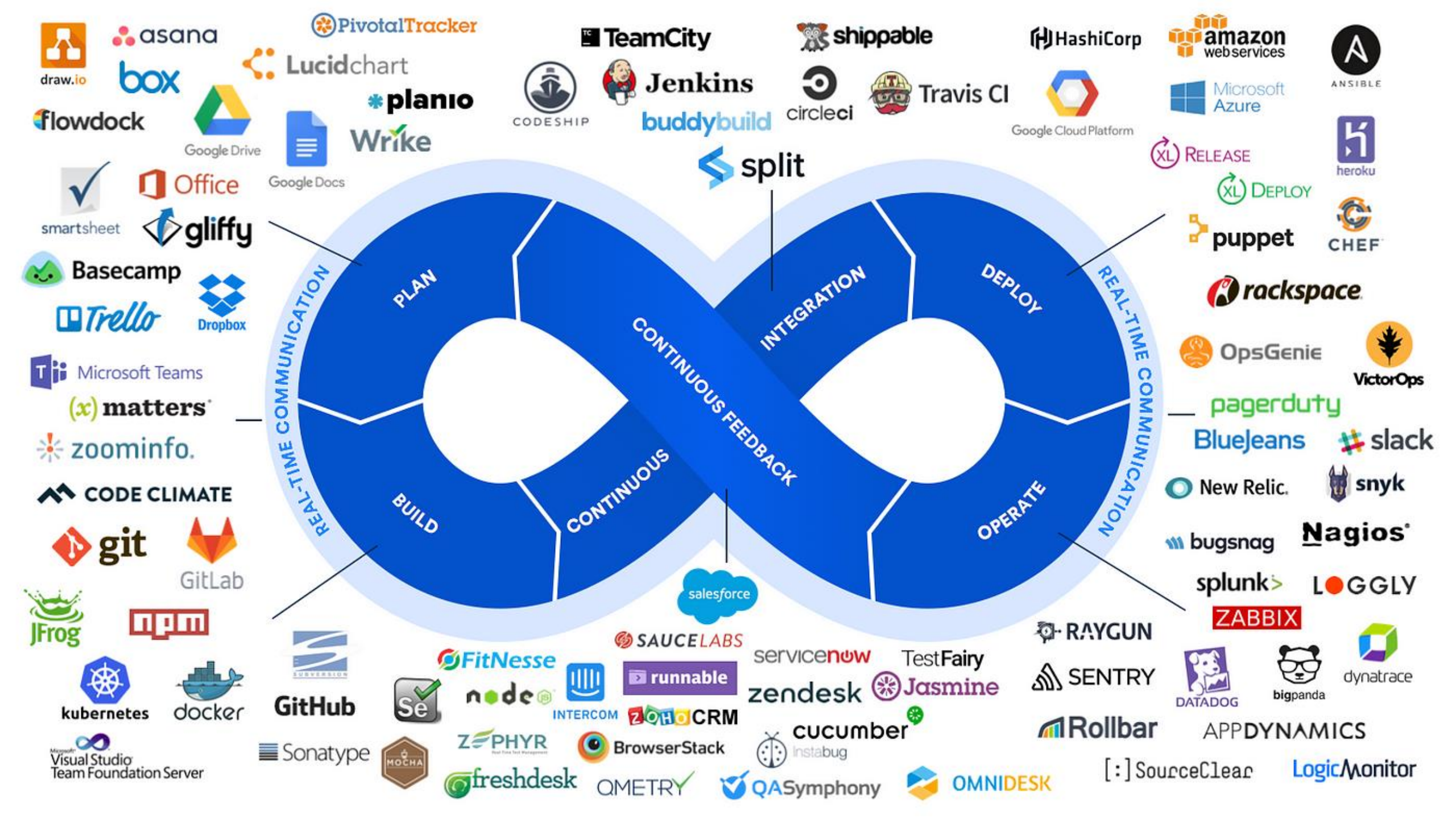


DevOps LifeCycle Phase



DevOps Tools


Confluence
JIRA
git
DC/OS
docker
aws
CHEF
sbt
maven
CODE
BUILD
TEST
PLAN
RELEASE
DEPLOY
OPERATE
MONITOR
ANSIBLE
kubernetes
Se
JUnit
Jenkins
CODESHIP
Nagios
splunk>
DATADOG


PLAN
CONTINUOUS
INTEGRATION
DEPLOY
OPERATE
BUILD
CONTINUOUS FEEDBACK
REAL-TIME COMMUNICATION
REAL-TIME COMMUNICATION
draw.io
asana
PivotalTracker
box
Lucidchart
planio
flowdock
Google Drive
Google Docs
Wrike
smartsheet
Office
gliffy
Basecamp
Trello
Dropbox
Microsoft Teams
(x) matters
zoominfo.
CODE CLIMATE
git
GitLab
JFrog
npm
kubernetes
docker
Visual Studio Team Foundation Server
GitHub
Sonatype
MOCHA
FitNesse
node
INTERCOM
runnable
ZOHO CRM
ZEPHYR
BrowserStack
freshdesk
QMETRY
SAUCELABS
salesforce
servicenow
zendesk
cucumber
Instabug
QASymphony
OMNIDESK
TestFairy
Jasmine
TeamCity
Jenkins
CODESHIP
buddybuild
shippable
circleci
Travis CI
split
HashiCorp
amazon web services
Google Cloud Platform
Microsoft Azure
ANSIBLE
RELEASE
DEPLOY
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puppet
CHEF
rackspace
OpsGenie
VictorOps
pagerduty
BlueJeans
slack
New Relic.
snyk
bugsnag
Nagios
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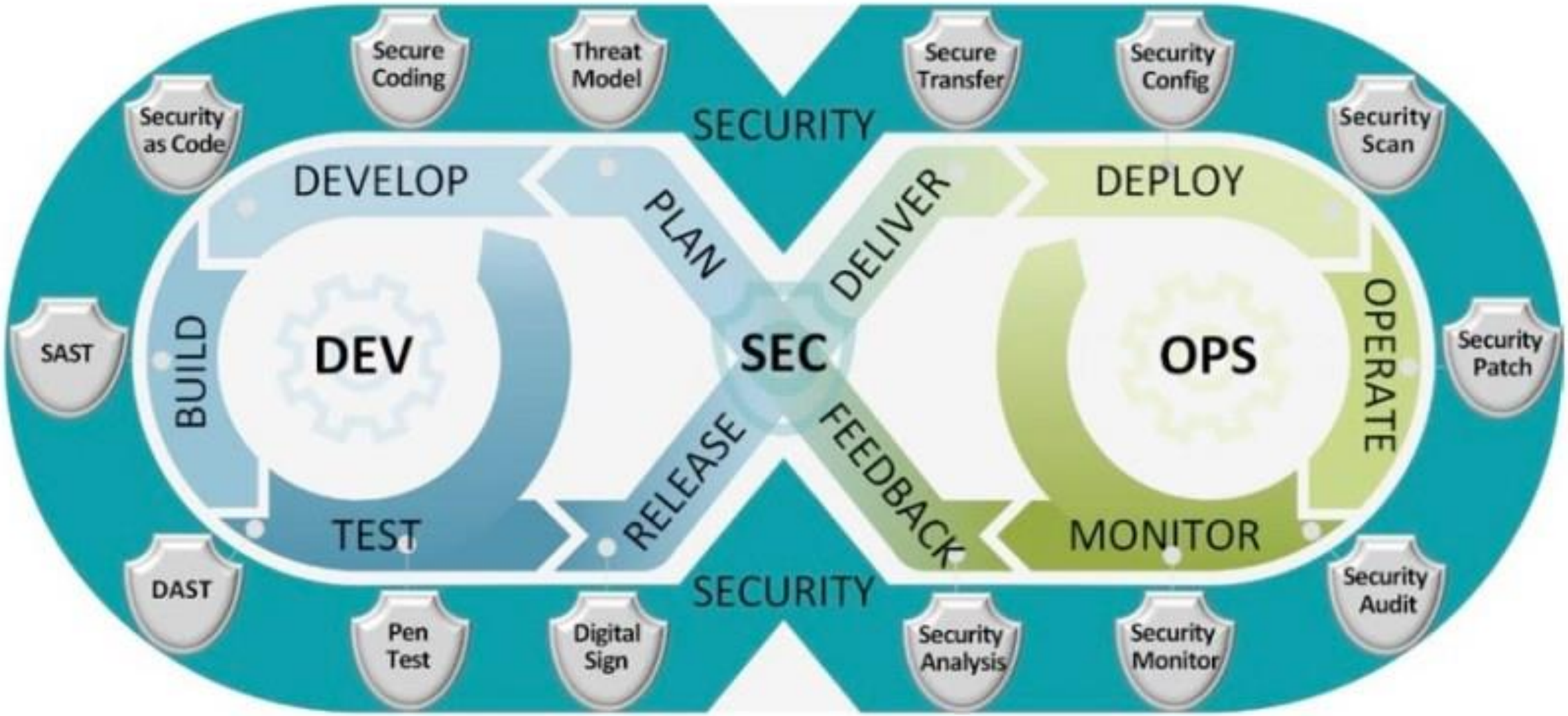
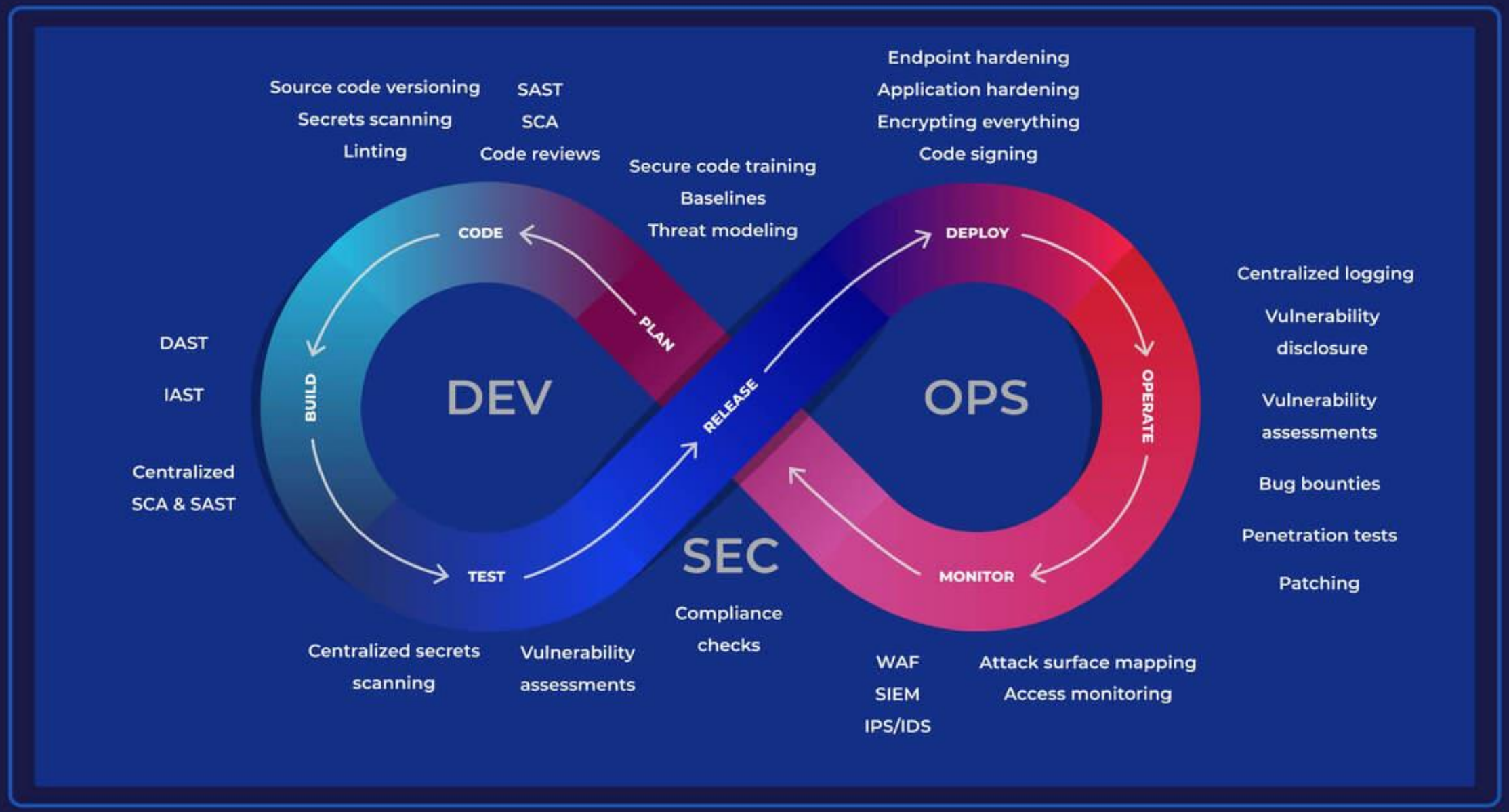
ZABBIX
RAYGUN
SENTRY
DATADOG
bigpanda
dynatrace
Rollbar
APPDYNAMICS
[:]SourceClear
LogicMonitor


Secure Coding
Threat Model
Secure Transfer
Security Config
Security as Code
Security Scan
SECURITY
DEVELOP
PLAN
DELIVER
DEPLOY
SAST
BUILD
DEV
SEC
OPS
OPERATE
Security Patch
RELEASE
FEEDBACK
TEST
MONITOR
DAST
SECURITY
Security Audit
Pen Test
Digital Sign
Security Analysis
Security Monitor


Source code versioning
Secrets scanning
Linting
SAST
SCA
Code reviews
Secure code training
Baselines
Threat modeling
Endpoint hardening
Application hardening
Encrypting everything
Code signing
CODE
PLAN
DEPLOY
DAST
IAST
Centralized
SCA & SAST
BUILD
DEV
RELEASE
OPS
OPERATE
Centralized logging
Vulnerability
disclosure
Vulnerability
assessments
Bug bounties
Penetration tests
Patching
TEST
SEC
MONITOR
Compliance
checks
Centralized secrets
scanning
Vulnerability
assessments
WAF
SIEM
IPS/IDS
Attack surface mapping
Access monitoring

spacelift

CONTINUOUS DEVELOPMENT

- This phase involves ‘planning‘ and ‘coding‘ of the software application’s functionality.
- The vision of the project is decided during the ‘planning’ phase.
- Code can be done in Any Language but maintain by Version Control System.

Why Code Versioning is Imp

- Versions are maintained to hold a single source of Application.
- Using Centralized single source code, Operations can access the same code what they plan to release.
- Easy to Rollout the faulty snippet of code or complete release.

CONTINUOUS TESTING

- This phase involves testing the code's of the software application's functionality.
- Testing could be done in Any Tool, Junit, Selenium…

CONTINUOUS INTEGRATION

- This is the most important phase of the DEVOPS lifecycle.
- All the Devops tools are very well integrated here.
- Scheduling of all the JOBS are done.
- Single point for the Developer and Operations teams to check on the error's and collaborate.

CONTINUOUS DEPLOYMENT

Configuration Management

Containerization

Configuration Management

- Establish & Maintain Application's functional requirements and Performance.
- Releasing deployments to Servers.
- Scheduling Updates on all Servers.
- Maintain Configuration Consistency on All Servers.

Containerization

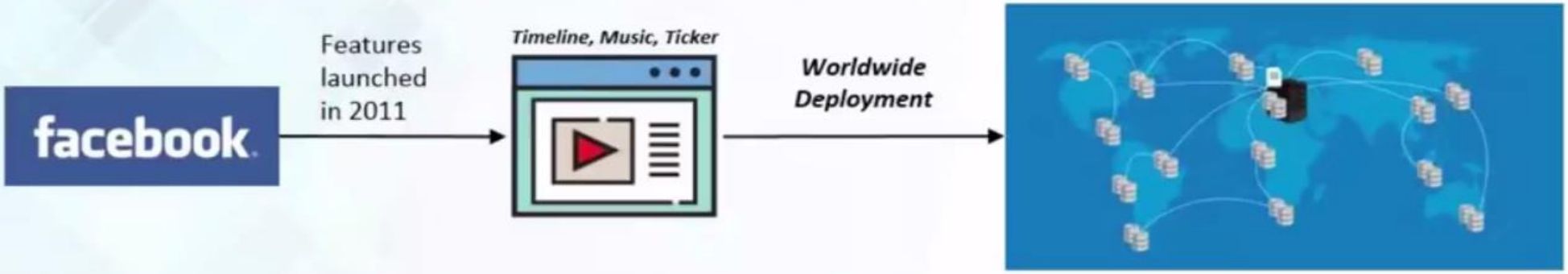
- Containerization tool is a set of Tools which will maintain the consistency across the environments.
- Docker is the First Containerization Tool.
- Scheduling Updates on all Servers.
- Maintain Configuration Consistency on All Servers.

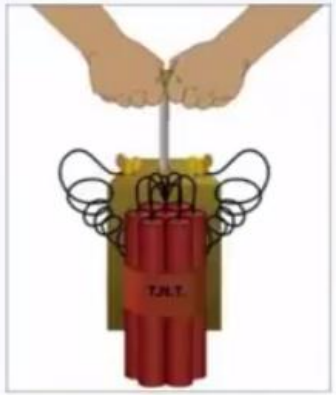
CONTINUOUS MONITORING

Continuous Monitoring is monitor the System Performance.

Monitor the Product overall Performance.

DEVOPS USECASE





Challenges they faced that day

- Features released to 500 million users → Heavy Website traffic → Server Meltdown
- Mixed responses from users which lead to no conclusion

Use Case 2011 : FaceBook

Post This They Came Up With

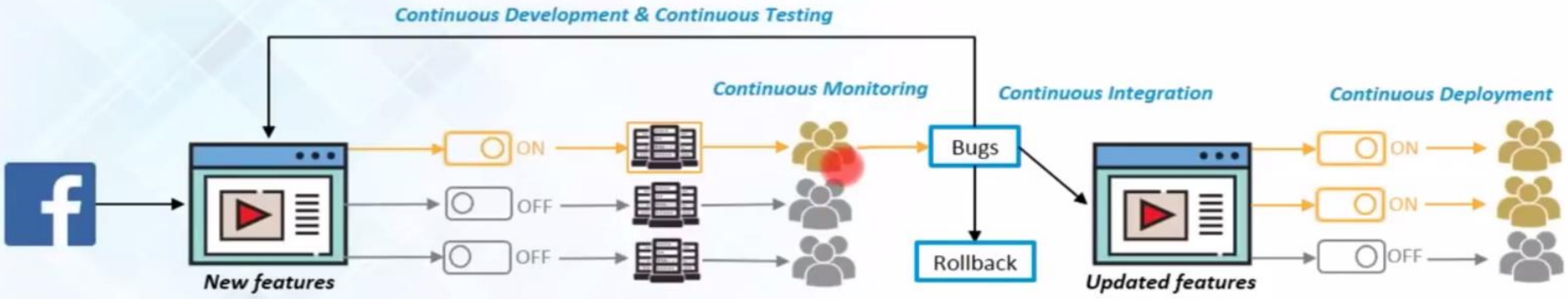
The Dark Launching Technique

Face Book Comes Up with Dark Launching

According to Dark Launching Technique:

- The new features are first deployed on a smaller & specific user base.
- They are continuously monitored and the feedbacks are continuously developed and tested.
- Once the features are stable, they are deployed on other user bases in multiple releases.

The Dark Launching Techinique



To Implement Dark launching, the below activities are fundamental as they lie at the heart of the DevOps lifecycle:

- Continuous Development
- Continuous Testing
- Continuous Integration
- Continuous Deployment
- Continuous Monitoring

The Dark Launching Techinique

PIPELINE OVERVIEW